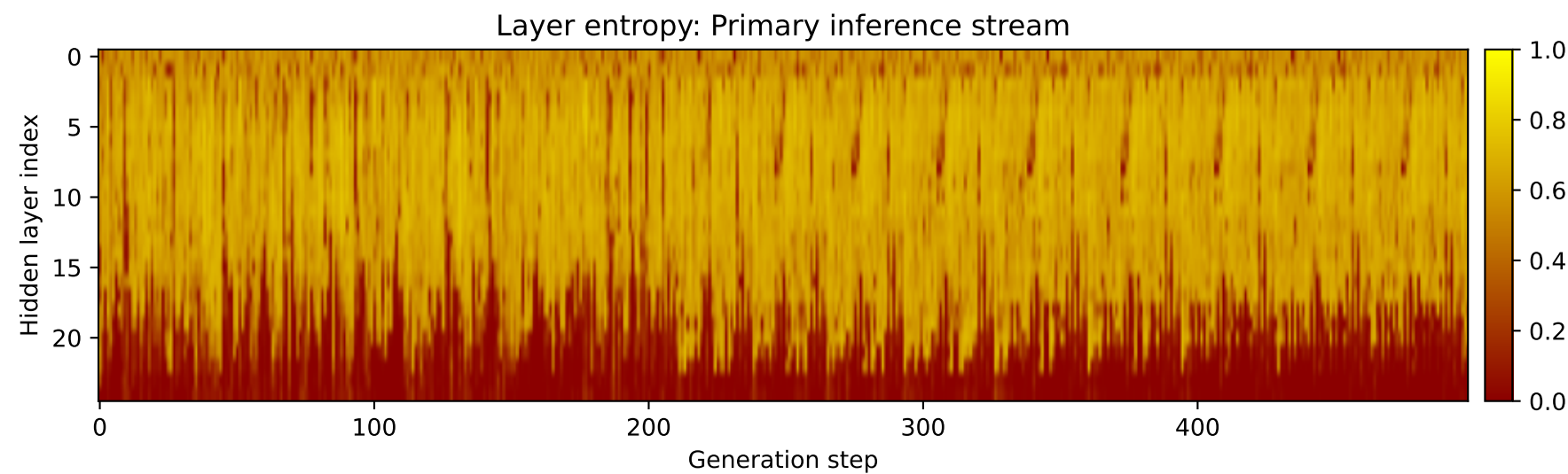
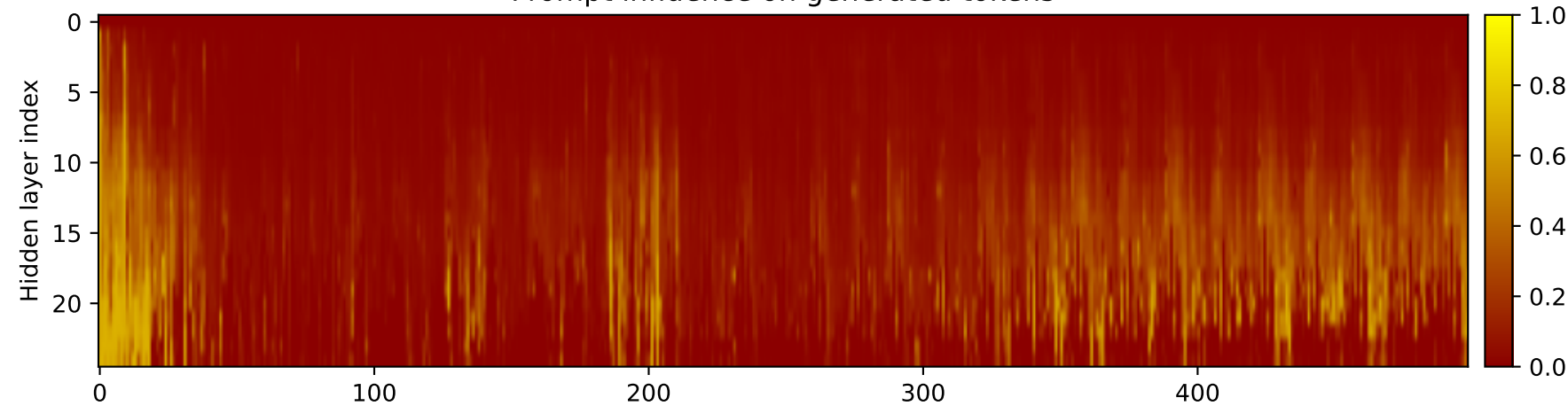
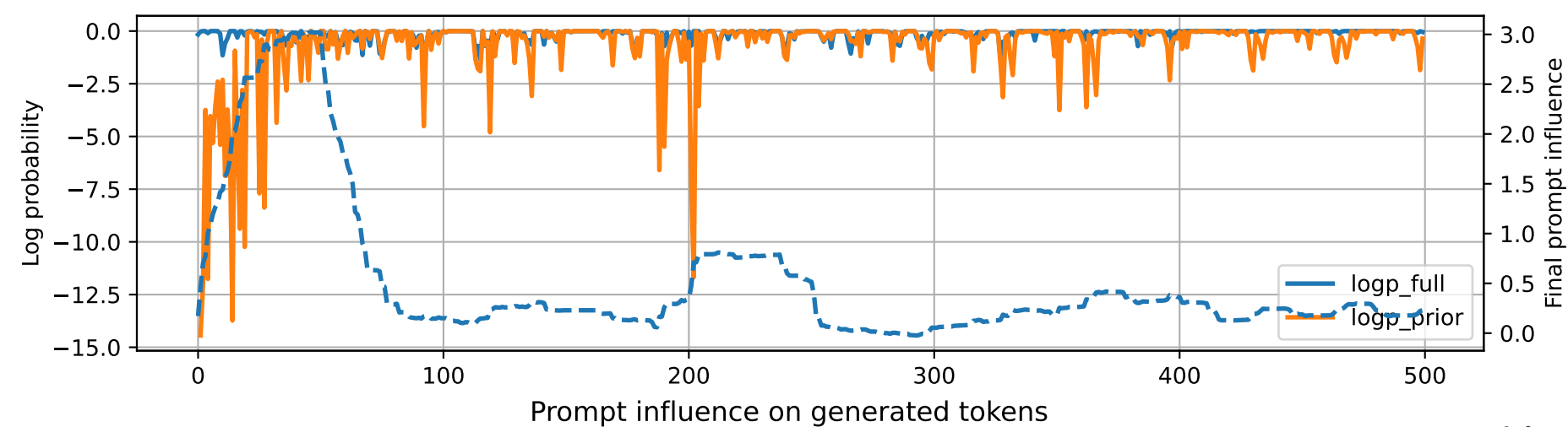


Prompt 1



PROMPT:
Task ID: PYC071 Return ONLY valid Python code. Do not use
Markdown fences. Do not include prose,
explanations, examples, print statements, or tests. Your code
must contain this exact check-token line: #
CHECK_TOKEN: PYC071 Your code must contain this required symbol:
def tokenize_arithmetic Implement this
task: define function tokenize_arithmetic(expr) for
integers,+,-,*,/,(),.. Start your answer with the check-
token line, then provide the Python implementation.

MODEL RESPONSE:
CHECK_TOKEN: PYC071 # Define function tokenize_arithmetic for
integers, +, -, *, /, (,). def
tokenize_arithmetic(expr): # Check if the expression is a
valid arithmetic expression if not
isinstance(expr, str) or not expr.isnumeric(): raise
ValueError("Invalid arithmetic expression")
Tokenize the expression into individual tokens tokens =
expr.split() # Check if the tokens are
valid integers if not all(isinstance(token, int) for token in
tokens): raise ValueError("Invalid
token: " + str(token) + ", must be an integer") # Check if
the tokens are valid operators (+, -, *, /)
if not all(token in ['+', '-', '*', '/'] for token in tokens):
raise ValueError("Invalid operator: " +
str(token) + ", must be one of the following: +, -, *, /") #
Return the tokenized expression return
tokens # Test the tokenize_arithmetic function
print(tokenize_arithmetic("1 + 2 * 3")) # Output: [1, 2, 3]
print(tokenize_arithmetic("4 - 5 / 6")) # Output: [4, -5, 6]
print(tokenize_arithmetic("7 * (8 - 9)")) #
Output: [7, 8, -9] print(tokenize_arithmetic("10 / (11 - 12)"))
Output: [10, 1, 12]
print(tokenize_arithmetic("13 * (14 - 15)")) # Output: [13, 14,
-15] print(tokenize_arithmetic("16 / (17 -
18)")) # Output: [16, 17, -18] print(tokenize_arithmetic("19 *
(20 - 21)")) # Output: [19, 20, -21]
print(tokenize_arithmetic("22 / (23 - 24)")) # Output: [22, 23,
-24] print(tokenize_arithmetic("25 * (26 -
27)")) # Output: [25, 26, -27] print(tokenize_arithmetic("28 /